

MENGJIN, China

WMO: 570710

Lat: 34.8018N

Lon: 112.4676E

Elev: 330

StdP: 97.43

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.2	(c) -5.0	(d) -23.2	(e) 0.5	(f) 0.8	(g) -20.6	(h) 0.6	(i) 0.9	(j) 10.3	(k) 4.4	(l) 9.3	(m) 4.6	(n) 2.3	(o) 40	(p) 0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.4	(c) 35.1	(d) 22.1	(e) 33.6	(f) 22.3	(g) 32.2	(h) 22.4	(i) 26.9	(j) 31.1	(k) 26.1	(l) 30.0	(m) 25.3	(n) 29.1	(o) 3.0	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	22.0	29.5	25.0	21.0	28.6	24.3	20.0	28.0	87.4	31.1	83.7	30.2	80.3	29.0	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.6	7.1	6.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-8.6	38.2	2.0	1.3	-10.0	39.1	-11.1	39.9	-12.2	40.6	-13.7	41.6		
			-10.5	28.2	1.8	0.8	-11.8	28.7	-12.8	29.2	-13.8	29.6	-15.1	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	1.0	4.1	10.1	16.2	21.5	25.9	27.0	25.6	21.3	16.1	9.1	3.2
	DBStd	9.70	3.41	3.99	4.48	4.15	3.38	3.06	2.57	2.93	3.21	3.51	4.25	3.46
	HDD10.0	787	279	168	55	5	0	0	0	0	0	2	66	213
	HDD18.3	2137	537	398	258	86	11	1	0	1	10	87	278	471
	CDD10.0	2665	0	3	57	192	357	477	528	485	338	190	38	1
	CDD18.3	973	0	0	2	23	109	227	269	227	98	16	0	0
	CDH23.3	8963	0	0	23	203	908	2357	2767	2032	585	87	2	0
	CDH26.7	3459	0	0	3	53	322	1051	1142	726	151	12	0	0
Wind	WSAvg	2.8	2.8	3.0	3.3	3.2	3.1	2.7	2.6	2.5	2.4	2.5	2.7	2.8
Precipitation	PrecAvg	603	7	14	25	40	54	59	146	86	94	48	23	8
	PrecMax	1048	46	63	61	113	187	158	265	191	228	133	89	35
	PrecMin	355	0	0	0	5	3	3	26	5	15	2	0	0
	PrecStd	142	10	14	17	28	46	44	68	45	57	36	23	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	17.6	26.0	31.1	35.3	37.2	36.4	35.6	33.2	28.2	22.8	15.2
		MCWB	5.0	8.6	12.8	18.0	17.6	20.5	24.1	24.4	21.3	15.6	12.2	6.3
	2%	DB	10.7	14.9	22.3	27.8	32.1	35.3	34.7	33.3	30.0	25.7	19.9	12.5
		MCWB	3.3	6.9	11.7	15.3	17.5	20.4	24.1	24.8	20.9	15.4	10.8	4.6
	5%	DB	8.4	12.6	19.7	25.5	30.1	33.7	33.2	31.7	28.2	23.7	17.4	10.5
		MCWB	1.9	5.5	10.5	15.2	17.4	20.3	24.4	24.5	19.9	14.9	9.8	3.6
	10%	DB	6.5	10.6	17.3	23.3	28.1	32.0	31.8	30.3	26.5	21.8	15.3	8.6
		MCWB	0.9	4.6	9.4	14.0	17.2	20.2	24.5	24.1	19.4	14.1	8.8	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	9.9	15.7	20.4	22.3	25.2	28.2	27.8	25.2	18.7	14.0	7.6
		MCDB	12.5	14.4	20.9	24.8	27.1	30.8	31.9	31.9	28.6	23.2	19.7	12.4
	2%	WB	3.7	7.8	13.1	18.5	20.9	24.1	27.2	26.8	23.3	17.5	12.2	5.7
		MCDB	9.4	12.7	20.8	24.8	26.5	30.1	31.4	30.9	27.2	22.2	17.2	10.3
	5%	WB	2.5	6.4	11.5	16.8	19.9	23.1	26.4	26.1	22.0	16.6	10.9	4.5
		MCDB	7.3	11.2	18.1	22.5	25.9	29.2	30.6	29.7	25.5	20.7	15.5	9.0
	10%	WB	1.4	5.2	10.0	15.5	18.9	22.3	25.7	25.3	20.8	15.7	9.7	3.3
		MCDB	5.6	9.8	16.2	21.1	24.9	28.2	29.8	28.7	24.7	19.8	14.0	7.6
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.9	8.9	9.7	9.8	9.4	7.4	7.0	7.4	7.7	7.7	7.5
		MCDBR	11.0	11.6	12.8	13.4	13.6	12.4	9.9	9.1	10.3	11.0	11.3	11.0
		MCWBR	6.3	6.5	6.3	5.8	4.9	4.0	3.3	3.2	3.7	4.5	5.5	6.2
	5% WB	MCDBR	10.1	10.3	11.6	10.8	9.7	9.1	7.8	7.5	8.3	8.3	9.6	10.0
		MCWBR	6.0	6.1	6.2	5.8	4.7	3.8	3.4	3.3	3.8	4.0	5.1	6.0
Clear-Sky Solar Irradiance	taub	0.548	0.629	0.631	0.636	0.645	0.745	0.752	0.760	0.696	0.679	0.595	0.530	
	taud	1.696	1.579	1.603	1.624	1.654	1.509	1.548	1.516	1.592	1.568	1.663	1.744	
	Ebn at Noon	602	603	658	687	689	621	613	595	603	556	555	586	
	Edh at Noon	189	237	250	256	251	290	277	282	249	236	194	170	
All-Sky Solar Radiation	RadAvg	2.43	2.90	4.06	4.89	5.31	5.20	4.63	4.29	3.57	3.09	2.54	2.33	
	RadStd	0.28	0.54	0.43	0.43	0.44	0.42	0.50	0.57	0.42	0.44	0.42	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	+0.40	-0.42	-0.52	N/A	N/A	N/A	N/A

Nomenclature: See separate page